



Armada Kalamunda Group

Cervical Spine Injury Guideline

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1. Purpose

To guide staff with clinical decision-making in assessment of cervical spine injuries in Adult and Paediatric patients.

2. Applicability

This guideline has been developed to guide clinical decision-making for the medical, nursing and allied health staff within the Armadale Kalamunda Group (AKG) Emergency Department.

3. Cervical Spine Injury Guidelines

All suspected or confirmed spinal injuries should be treated as unstable until proven otherwise. It is important to maintain spinal precautions until documented spinal clearance, or directives for further management given by Spinal Team (RPH/PCH).

See Appendix for procedure for [Cervical neck hold](#), [Log rolling](#) and [Full spinal precautions](#)

3.1 Adult C-Spine Injury

Refer to:

- RPH [Spinal Clearance Guideline](#)
- AKG [Management of Patients with a Spinal Injury Procedure](#)

3.1.1 Evidence-based decision rules for evaluating the cervical spine after trauma

Either decision rule may be used

- [Canadian C-Spine Rule](#)
- [NEXUS Criteria](#)

3.1.2 Diagnostic Imaging Pathway

- [Cervical Spine Trauma Diagnostic Imaging Pathway](#)

3.2 Paediatric C-Spine Injury

Refer to:

- [PCH Emergency Department Guidelines - Cervical Spine Trauma](#)

3.2.1 Evidence-based decision rule for evaluating the cervical spine after trauma

- [PECARN C-Spine Imaging Rule](#)

3.2.2 Diagnostic Imaging Pathway

- [PCH C-Spine Clearance Tool](#)

3.3 Spinal Clearance

Spinal Clearance may only be performed by an ED Consultant or Registrar in Armadale ED, and must be clearly documented in the patient's medical record

- Clinical clearance may be performed using:
 - **Adults:** [NEXUS Criteria](#) or [Canadian C-Spine Rule](#)
 - **Paediatrics:** [PCH C-Spine Clearance Tool](#)
- Radiological clearance:
 - Note: a radiological clearance on plain XR or CT does not exclude risk of ligamentous injury

3.4 Confirmed/Suspected C-Spine Injury

Following assessment C-spine Injury may be:

- **Confirmed** on imaging, or
- **Suspected** despite normal imaging e.g.
 - alert patients with ongoing neurological symptoms or
 - unevaluable obtunded patients with significant mechanism
- **Unlikely**, spinal clearance needs to be documented in medical records

If confirmed/suspected C-spine injury, maintain C-spine precautions and refer patient:

- **Isolated C-spine Injury:** Discuss with the **Spinal Team at RPH/PCH**
- **Multi-trauma:** Refer to **RPH Trauma Service/PCH Surgical Team**

4. Legislative context

The following documents are required to give effect to this policy:

- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0171/22](#)
- HDWA [Consent to Treatment Policy MP 0175/22](#)

5. Supporting information

The following documents support the implementation of this policy:

- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-22](#)
- [Management of Patients with a Spinal Injury Procedure AKG-PRO-0627-24](#)

6. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the departmental meeting for review and identification of any opportunities for improvement.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 8 – Action 8.04 Recognising acute deterioration
- Standard 8 – Action 8.06 Escalating care

8. References

1. RPH Spinal Clearance Guideline: <https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/EMHS/RPH/Spinal%20Clearance%20Clinical%20Guideline.pdf>
2. Radiology Across Borders Diagnostic Imaging Pathways: https://radiologyacrossborders.org/diagnostic_imaging_pathways/index.php/imaging-pathways/musculoskeletal-trauma/bone-and-joint-trauma/cervical-spine-injury?tab=nexus#pathway
3. PCH ED Cervical Spine Trauma Guideline: <https://pch.health.wa.gov.au/For-health-professionals/Emergency-Department-Guidelines/Cervical-spine-trauma>
4. PECARN prediction rule for cervical spine imaging of children presenting to the emergency department with blunt trauma: a multicentre prospective observational study. Leonard, Julie C et al. The Lancet Child & Adolescent Health, Volume 8, Issue 7, 482 – 490. [https://doi.org/10.1016/S2352-4642\(24\)00104-4](https://doi.org/10.1016/S2352-4642(24)00104-4)
5. RPH CPS for Acute Spinal Injury: <https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/EMHS/RPH/S%20-%20Spinal%20Patient%20Acute%20Injury%20CPS.pdf>
6. Khan AD, Liebscher SC, Reiser HC, Schroepel TJ, Anstadt MJ, Bosarge PL, Carroll SL, Quick JA, Barnes SL, Sobrino J, Murry J, Morin N, Gomez M, Consani H, Gonzalez RP. Clearing the cervical spine in patients with distracting injuries: An AAST multi-institutional trial. J Trauma Acute Care Surg. 2019 Jan;86(1):28-35. <https://doi.org/10.1097/TA.0000000000002063>

9. Document contact

Enquiries relating to this guideline may be directed to:

Title: Dr Emma Trace, Consultant Emergency Physician
 Department: Emergency Department
 Email: emma.trace@health.wa.gov.au

10. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V1	10/12/2024	10/12/2027	Document separated from the original C-Spine and C-Head Injury Guideline

11. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Tania Strickland – A/Director Clinical Services
Authorisation date	10/12/2024
Published date	10/12/2024

Appendix 1: Cervical neck hold

The person who performs the cervical neck hold is the team leader and responsible for directing and managing the turn from beginning to end. The team leader must have the required skills, knowledge, and experience to perform a cervical neck hold

- A minimum of four staff are required (1 for the cervical neck hold, 2 to reposition the patient and 1 to perform procedures)
- Assess the need for pain relief and administer prior to procedure
- Explain the procedure to the patient and the turning team
- Adjust the bed height according to manual handling principles
- Remove any lateral towel bolsters before rolling

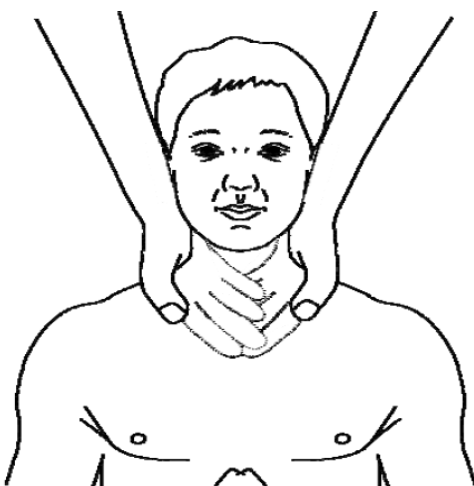


For supine to lateral positioning:

- Place the fingers of one hand at the base of the patient's neck and the thumb over the clavicle
- Hold the arm along the length of the head at the height of the ear
- Place the other hand and arm in the same position as above crossing fingers over
- Hands are in the position shown below

For lateral to supine positioning:

- Place the fingers of the hand closest to the bed around the patient's neck and the thumb over the clavicle
- Hold the arm along the length of the face



Appendix 2: Log rolling

The log rolling technique is utilised until the patient no longer requires “spinal precautions”

- A minimum of 3 staff are required for log rolling an adult patient
 - If a [cervical neck hold](#) is required, an extra staff member is needed
 - Additional staff may be required for ventilated patients, bariatric patients or those with other injuries
- Assess the need for pain relief and administer prior to performing procedure
- Explain the procedure to the patient and the turning team
- Adjust the bed height according to manual handling principles
- Remove any objects which may obstruct the turn and ensure any lines will not be compromised by moving the patient
- The person closest to the patient’s head is always responsible for the count and instructs the team (Team Leader)
- The Team Leader decides which direction patient to be rolled, considering the nature and location of any injuries sustained by patient
- Team members stand on side of bed that patient is to be rolled to face
- Prepare to roll by placing hands on furthest side of patient:
 - **1st person** (tallest member of team) - on the shoulder and hip
 - **2nd person** (middle height) - on the hip and under thigh
 - **3rd person** (shortest height) - under knee and lower leg



- The team leader will state on the count of three to roll the patient onto their side in a smooth coordinated move, avoiding any rotational movement of the spine
- **3rd person** to support weight of leg and keep lateral malleolus in line with the hip during turn to prevent adduction of leg
- Perform assessment/treatment required
- Then Team leader will count to three to roll the patient back onto their back
- Position lines, lateral bolsters, support pillows etc as necessary
 - If neurological deficit is present: - place pillows at the end of the bed to ensure both feet are flexed at 90 degrees
- Ensure spinal alignment is correct:
 - Patient supine: centre of forehead, chin, xiphoid sternum and pubic bone in line centrally. The shoulders and pelvic crests should not be rotated
- Ensure the patient has the call bell and other necessary items before leaving bedside

Appendix 3: Full spinal precautions (FSP)

FSP are the maintenance of spinal alignment, at all times. FSP are to be implemented for any movement or transfer of a patient with a diagnosed or suspected spinal injury until otherwise documented in the patient health records by a MO

All staff involved with moving patients with suspected or confirmed vertebral column or spinal injuries must have completed education in manual handling

- The patient must be positioned on an appropriate standard hospital mattress when in bed on FSP (**no air mattresses are permitted**)
- No pillows are to be placed under the patient's head, or any other part of their body, unless prescribed by the MO. If supports are going to be used under the patient's body the exact location is to be documented in the patient health record by MO

In patients with a fixed flexion/brittle bone deformity such as that seen in Ankylosing Spondylitis, spinal precautions need to be tailored to maintain the patient's pre-injury kyphotic curvature to prevent hyperextension. This can be achieved using lateral bolsters, pillows and foam/rolled blankets

- The hard extrication collar should be used for a maximum of six (6) hours ONLY from placement by SJA. It should be exchanged for a [Philadelphia® collar](#) as soon as possible, and should be applied by a skilled staff member. Miami J™ collars are to be fitted and adjusted by an Orthotist
- If a collar needs to be removed for any reason before [spinal clearance](#) is documented, a [cervical neck hold](#) is to be instituted by a skilled staff member
- Whenever a collar comes off skin assessment & care must occur
- While every effort should be made to maintain neutral positioning in high-risk suspected injury, collars or other immobilisation techniques should never be forced on a patient. Early senior doctor input is mandatory
- Educate patient not to move head, neck or back
- A designated and skilled team leader (TL) coordinates all movements of this patient. All others involved in patient movement wait for direction from the TL
- For any movement immobilise the patient's cervical spine via a [neck hold](#) and [log roll](#) them – if c-spine has been documented as cleared, cervical neck hold no longer required
- Skilled staff need to have communication of the move (e.g. "ready, steady, go or 1,2,3")
- When the TL says "go or 3" the patient is rolled. No one else is to take on this role
- The patient on FSP must be lying flat in the bed. Some patients will be allowed to have their head of bed elevated for example, "to 30". Head of bed elevation must be clearly written in the patient health records by MO. All other spinal precautions still apply
- Anatomical alignment of the patient must always be maintained. This includes during the process of moving the patient. The TL is responsible for checking and maintaining the spinal alignment
- Prescribed orthoses are to be worn at all times unless clearly documented otherwise in the patient health records by MO
- The injury site is supported/immobilised as follows:
 - Cervical injuries – by the prescribed collar (usually a [Philadelphia®](#) or Miami J™), a [cervical neck hold](#) is instituted during [log rolls](#), lifts, and transfers

- Thoracic/lumbar injuries – are supported by placement of hands anteroposteriorly (front and back) at the level of the injury during rolls. Thoraco/lumbar orthoses are worn for ambulation and put on the patient before they get out of bed
- Skilled staff are required to escort patients with a suspected or diagnosed cervical, thoracic or lumbar vertebral fracture or cord injury when the patient is transferred to other areas within the hospital, e.g. X-ray or Operating Rooms. Suction MUST accompany all transfers
- When the patient is to be transferred to RPH a Paramedic Crew Ambulance should be booked. At the time of booking the Paramedic Crew Ambulance, please advise of the need for the Ferno Vacuum Mattress. The patient may still require Medical or Nurse Escort. In some cases, both will be required. If unsure, discuss your concerns with and attain guidance from the ED DMO
- Consideration needs to occur regarding the patient's needs prior to transporting to another hospital, e.g. antiemetic's should be prescribed and administered where appropriate, ensure bladder care has occurred

Appendix 4: Philadelphia ® adjustable sizing and fitting guide

PHILADELPHIA® ADJUSTABLE SIZING & FITTING GUIDE

HEIGHT ADJUSTMENT



NECK HEIGHT

With spine in desired alignment, place the chin-cup of the front to the patient's chin and slide the lower front down to the patient's sternum (Figure 1). The superior border of the sternal pad should sit just under the sternal notch. Lift the front away from the patient and push the two tabs to lock the size (Figure 2).

SIZE ADJUSTMENT



CIRCUMFERENCE

After applying the back piece of the collar, tear off any sections anterior of the earlobe (Figure 3 and 4). If unsure, hold front piece up to chin and ensure there is adequate crossover of both sections to allow a secure fit.

Warning: This collar is designed to be sized and locked into a single position only once; it is not intended to be resized.

Note: If you are familiar with the previous Philadelphia sizes you can use the size numbers marked on the front parts where 2 refers to 2 1/4", 3 refers to 3 1/4", etc. Simply slide the lower front down until the correct size number appears in the size window and lock.

APPLICATION



1. After adjusting the collar, apply the back piece of the collar to the back of the patient's neck. Center the collar. The back arrow should point upward.

2. Apply the front piece of the collar with the chin centered in the recess. The FRONT piece OVERLAPS the back piece to ensure effective immobilisation and comfort. The front arrow should point upward.

3. With Velcro® hook and loop fasteners, tighten the collar with a bilateral adjustment. Check that spine is in desired alignment.



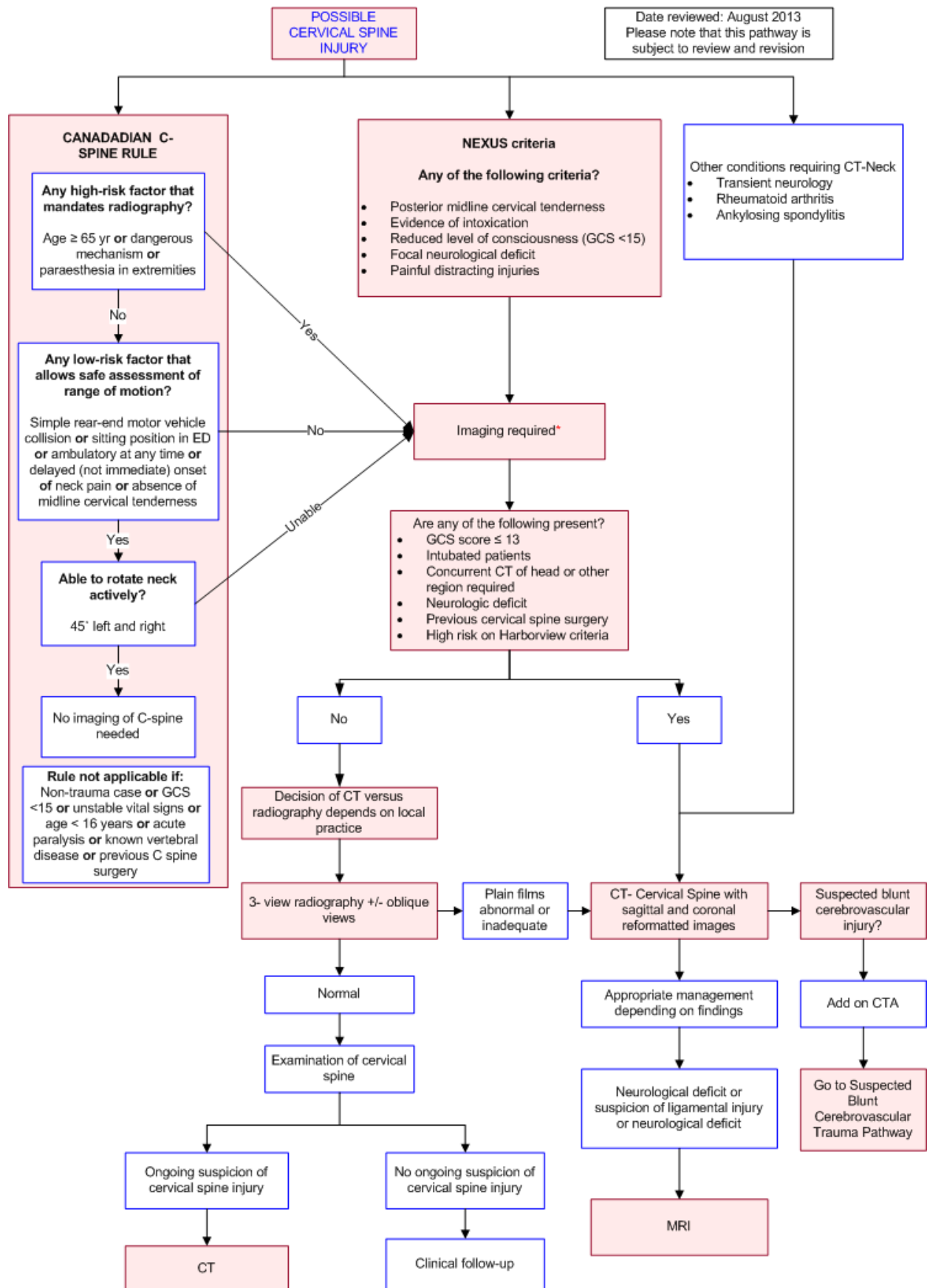
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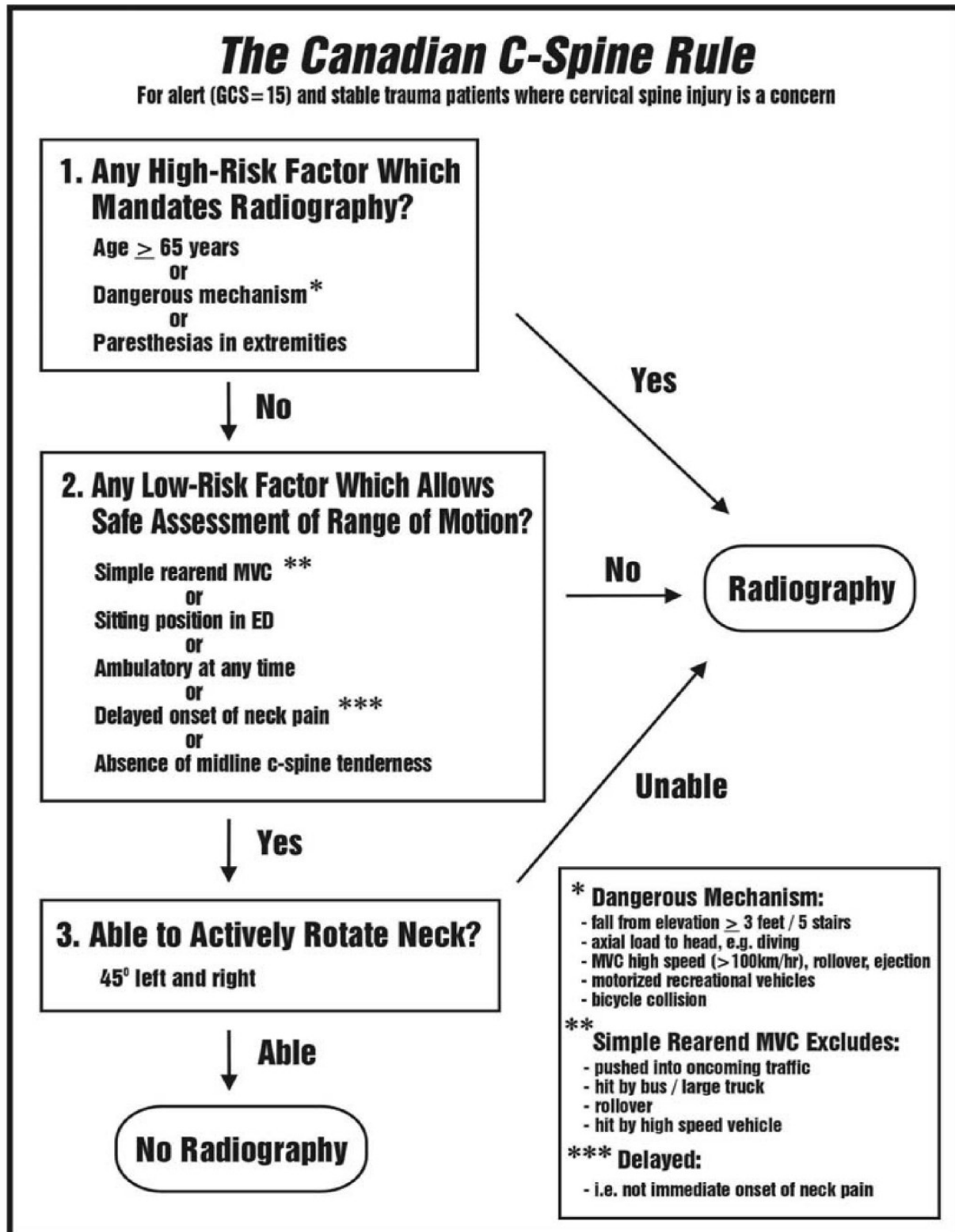


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Appendix 5: Cervical Spine Diagnostic Imaging Pathway



Appendix 6: Canadian C-Spine Rule



Appendix 7: NEXUS Criteria

NEXUS Criteria:	
• No Focal Neurological Deficit • No Midline Spinal Tenderness • No Altered Level of Consciousness • No Intoxication • No Distracting Injury	If ALL criteria are met, the C-Spine can be cleared clinically and imaging is not required

The NEXUS Low Risk Criteria:

- Cervical spine radiography is indicated for patients with trauma unless they meet **all** the following criteria:

No posterior midline cervical spine tenderness

- Midline posterior bony cervical spine tenderness is present if the patient reports pain on palpation of the posterior midline neck from the nuchal ridge to the prominence of the first thoracic vertebra, or if the patient evinces pain with direct palpation of any cervical spinous process

No evidence of intoxication

- Patients should be considered intoxicated if they have either of the following: a recent history provided by the patient or an observer of intoxication or intoxicating ingestion, or evidence of intoxication on physical examination such as an odour of alcohol, slurred speech, ataxia, dysmetria or other cerebellar findings, or any behaviour consistent with intoxication
- Patients may also be considered intoxicated if tests of bodily secretions are positive for alcohol or drugs that affect the level of alertness

A normal level of alertness

- An altered level of alertness can include any of the following: a Glasgow Coma Scale score of 14 or less; disorientation to person, place, time or events; an inability to remember three objects at five minutes; a delayed or inappropriate response to external stimuli; or other findings

No focal neurologic deficit

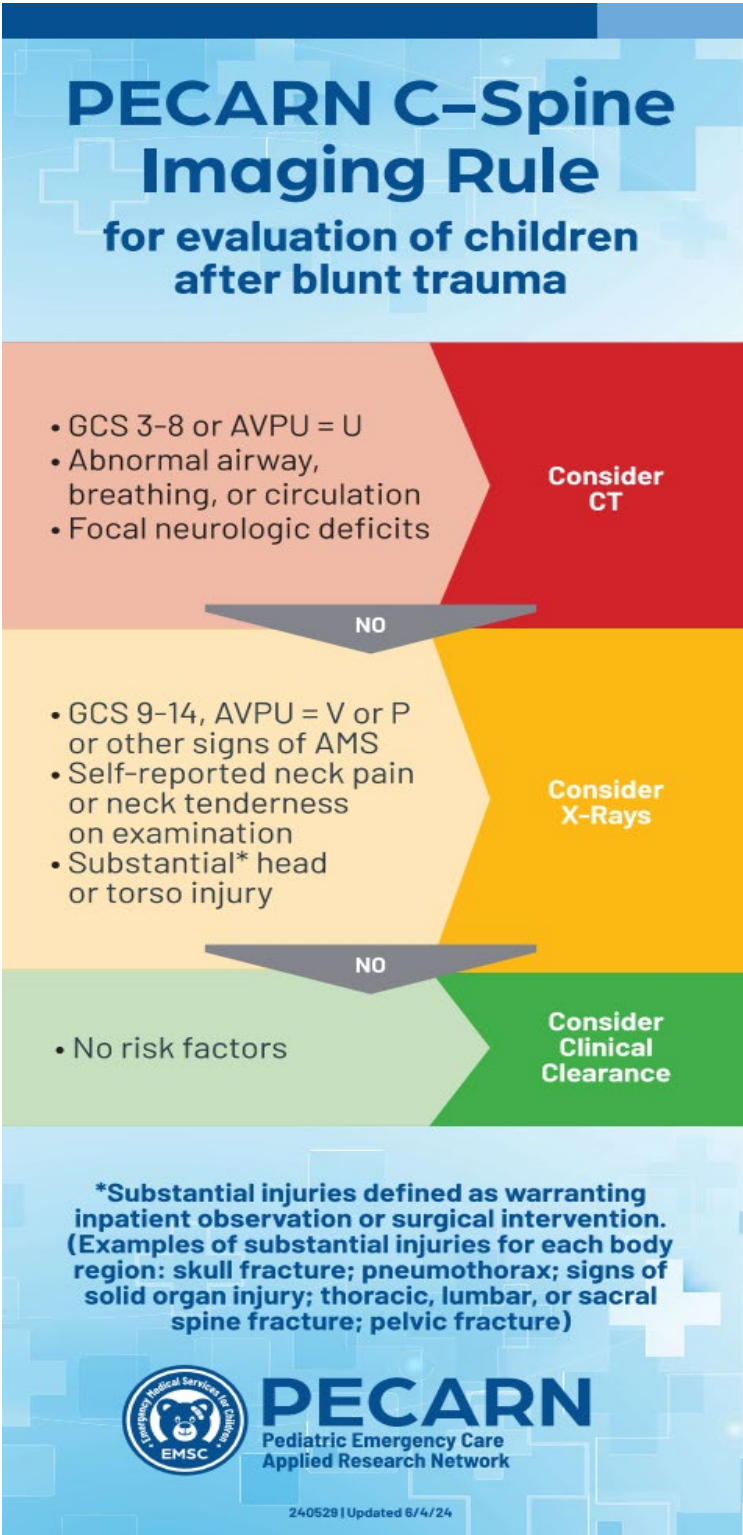
- A focal neurologic deficit is any focal neurologic finding on motor or sensory examination

No painful distracting injuries

- No precise definition of a painful distracting injury is possible
- This category includes any condition thought by the clinician to be producing pain sufficient to distract the patient from a second (neck) injury
- Such injuries may include, but are not limited to any long bone fracture, a visceral injury requiring surgical consultation, a large laceration, degloving injury or crush injury, large burns, or any other injury causing acute functional impairment

- Physicians may also classify any injury as distracting if it is thought to have the potential to impair the patient's ability to appreciate other injuries
- A 2019 multi-centre study showed there appeared to be no difference in rate of missed injury when comparing patients with distracting injuries to those without, suggesting clinical examination alone may be sufficient to clear a C-spine ([Clearing the cervical spine in patients with distracting injuries: An AAST multi-institutional trial - PubMed \(nih.gov\)](#))

Appendix 8: PECARN C-Spine Imaging Rule



Appendix 9: PCH C-Spine Clearance Tool

